• Laplace oper-r: $\Delta u = \underline{\nabla} \cdot \underline{\nabla} u = \frac{\partial^2 u}{\partial x_1^2} + \dots + \frac{\partial^2 u}{\partial x_n^2}$

• Defn (Harmonic): u is harmonic in the dom-n Ω, if ① $u \in C^2(\Omega)$ & satisfies the Laplace eq-n $\Delta u = 0$ in Ω.

• Thm 7.1 (W/S) Recall Analytic f-ns C^∞

If $u \in C^0(\Omega)$ & $\Delta u = 0$, then u is <u>analytic</u> ($\exists$ Taylor exp. $\to$ f-n u.)
(So Harmonic f-ns are analytic & so smooth $\in C^\infty$).

$C^0(\Omega) = \{f(x) : f(x)$ is cont-s over $\Omega\}$

$C^1(\Omega) = \{f(x) : f(x) \in C^0(\Omega),\ f_{x_i} \in C^0(\Omega)\}$

$C^2(\Omega) = \{f(x) : f(x) \in C^0(\Omega),\ f_{x_i} \in C^0(\Omega),\ f_{x_i x_j} \in C^0(\Omega)\} \dots$

Taylor Series: Notation:

$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n)$ Symbol

$$D^\alpha f := D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n} f = \frac{\partial^{\alpha_1 + \alpha_2 + \dots + \alpha_n} f}{\partial x_1^{\alpha_1} \cdot \partial x_2^{\alpha_2} \cdots \partial x_n^{\alpha_n}}$$

wrt 2nd indep var.

$x^\alpha := x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$

$\alpha! := \alpha_1! \cdot \alpha_2! \cdots \alpha_n!$

$|\alpha| := \alpha_1 + \dots \alpha_n$

$$T_n(\underline{x}) = \sum_{0 \le |\alpha| \le n} \frac{D^\alpha f(\underline{x}^0)}{\alpha!} (\underline{x} - \underline{x}^0)^\alpha$$

Taylor Series

$T_{cc} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Higher Order PDE:

(1) $P(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha D^\alpha$ = Linear operator (1)

(2) $P_m(\underline{x}, D) = \sum_{|\alpha| \le m} a^\alpha(\underline{x}) D^\alpha$ = Principal part of Lin. op. (1).

(3) $P_m(\underline{x}, \underline{\xi}) = \sum_{|\alpha| \le m} a^\alpha(\underline{x}) \xi^\alpha$ = Principal symbol.

Idea: Info of sol-n of PDE is spread throug a surface $\perp$ $\underline{\xi}$. = <u>Characteristic direction</u>
(3) = 0 i.e. $P_m(\underline{x}, \underline{\xi}) = 0$ solve for $\underline{\xi}$. Find vectors $\langle v, \underline{\xi} \rangle = 0$ for this surface.

Given $a u_{xx} + b u_{xy} + c u_{yy} + \dots = 0$. $\Delta = b^2 - 4ac$ with $J \neq 0$

$\xi = \xi(x,y)$, $\eta = \eta(x,y)$ $\quad A u_{\xi\xi} + B u_{\xi\eta} + C u_{\eta\eta} + \dots = 0$ & $\Delta' = (B^2 - 4AC) = \Delta \cdot J^2$. &

$u_{xy} = u_{\xi\xi} \xi_x \xi_y + u_{\xi\eta} (\xi_x \eta_y + \xi_y \eta_x) + u_{\eta\eta} \eta_x \eta_y + \dots$

In all the coef-s a, b, c = const, the coord. f (ξ, η) might be different at every pt in Ω. Also $\Delta > 0$ = hyp, $\Delta = 0$ (par), $\Delta < 0$ Ellip.

Laplace eq-n. $\Delta u = \underline{\nabla} \cdot \underline{\nabla} u = 0 \to u_{xx} + u_{yy} + \dots = 0$

Spherical: $u(r, \theta, \varphi)$ $\quad \Delta u = \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \Lambda_3 u = 0$ where oper-r Λ_3 is

$$\Lambda_3 u = \frac{1}{\sin\varphi} \cdot \frac{\partial}{\partial \varphi}\left[\sin\varphi \cdot \frac{\partial u}{\partial \varphi}\right] + \frac{1}{\sin^2(\varphi)} \cdot \frac{\partial^2 u}{\partial \theta^2}$$

Polar: $\Delta u = \frac{1}{r} \cdot \frac{\partial}{\partial r}\left[r \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \cdot \frac{\partial^2 u}{\partial \theta^2} = 0$

Calc III.

a) Diverg. thm.

$$\underbrace{\int_\Omega \underline{\nabla} \cdot \underline{v}\, d\underline{x}}_{\text{vol. S}} = \underbrace{\int_{\partial\Omega} \underline{v} \cdot \underline{n}\, ds}_{\text{surf. S.}}$$

{ Ω is regular dom if Ω is open/conn (dom), bounded, has Lipschitz boundary (finit # of corners)

1) Green's first Id-ty:

$$\int_\Omega u \cdot \underline{\nabla}^2 w\, d\underline{x} = \int_{\partial\Omega} u \cdot \frac{\partial w}{\partial \underline{n}}\, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w\, d\underline{x}$$

(← Integration by parts)

2). Green's 2nd id-y:

$$\int_\Omega (u\, \underline{\nabla}^2 w - w \cdot \underline{\nabla}^2 u)\, d\underline{x} = \int_{\partial\Omega} \left(u \frac{\partial w}{\partial \underline{n}} - w \frac{\partial u}{\partial \underline{n}}\right) ds.$$

Heat eq-n: $\rho c \cdot u_t - \underline{\nabla}(k \cdot \underline{\nabla} u) = 0 \overset{k = \text{const.}}{\Rightarrow} \frac{\rho c}{k} \cdot u_t - \underline{\nabla} \cdot (\underline{\nabla} u) = 0$ & $t' = \frac{k}{\rho c} t$

$u(\underline{x}, t) = ?$ Addit. info needed: must be given.

IC: $u(x, y, z, 0) = \varphi(x, y, z)$ $\forall x, y, z \in \Omega$, $\boxed{u \in C^2(\Omega) \cap C^1(\bar{\Omega})}$

BC: What's happening on $\partial\Omega$

① Dirichlet BC: $u(x, y, z; t) = f(x, y, z; t)$ $\forall (x, y, z) \in \partial\Omega$, $t > 0$

② Neuman BC: $\frac{\partial u}{\partial \underline{n}}(x, y, z; t) = g(x, y, z; t)$ $\forall (x, y, z) \in \partial\Omega$, $t > 0$.

③ Mixed $\alpha \frac{\partial u}{\partial \underline{n}} + \beta u = h(x, y, z; t)$ (needed if convective BC)

§6.5. Well-Posed: ① sol-n $\exists$ ② sol-n unique ③ $|f - \tilde{f}(x,y)| < \delta \Rightarrow |u - \tilde{u}| < \varepsilon(\delta)$

§7.2 Sep-n of var-s: $u(r, \theta) = R(r) \cdot \Theta(\theta)$

§7.3. Preserving Har-ty: ① Trs $x' = x + x_0$, $y' = y + y_0$, ② Rotations, ③ Reflections $x' = x$, $y' = -y$, ④ Stretch $x' = \lambda x$, $y' = \dots$

§7.5. Prop-s of Laplace eq-n $\Delta u = 0$

Prop ② of Har. u: Thm MVP $u(\underline{r}_0) = \frac{1}{|S(\underline{r}_0, \delta)|} \iint_{S(\underline{r}_0, \delta)} u(\underline{r})\, ds$, $\forall r_0 \in \Omega$. surface

Prop ③ of harm. u: Thm (Max. Pr): u attains max/min on $\partial\Omega$.

D'Alambert sol-n $u = F(x+ct) + g(x-ct)$ $\quad u(x,t) = \frac{1}{2}(\varphi(x+ct) + \varphi(x-ct)) + \frac{1}{2} \int_{x-ct}^{x+ct} \psi(\tau)\, d\tau$

eumann prm: $\Delta u = 0$ in Ω, $\theta \in [-\pi, \pi]$

$\frac{\partial u}{\partial n}(r = r_0, \theta) = f(\theta)$ on $\partial\Omega$. $f \in C^0(\partial\Omega)$, For a sol-n to exist $\int_{\partial\Omega} f\, ds = 0$, $\int_{-\pi}^{\pi} f(\theta)\, d\theta = 0$. Then $A_0 = 0$ &

Look for the sol-n $u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} \left[A_n r^n \cos(n\theta) + B_n r^n \sin(n\theta)\right]$

& $f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\theta) + b_n \sin(n\theta)\right]$, $\Rightarrow$ find a_n, b_n $\longrightarrow A_n = a_n/n$, $\longrightarrow B_n = b_n/n$ for

$\boxed{a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\, d\theta,\ n = 0, 1, 2, \ldots}$ (i) (ii) Oct. 28.

Def-n (Ch.7, HW10) (Harmonic)

A f-n, u, is called harmonic in Ω, if $u \in C^2(\Omega)$ & satisfies $\Delta u = 0$.

Thm 7.1 (i) (ii)

If $u \in C^0(\Omega)$ & $\Delta u = 0$, then u is analytic. ($T_n(x) \to u$.) C^∞ (An.f.)

- HW10. (Change of indep. var-s $x' = x + a$, $y' = y + b$, don't forget Cauchy IVP

(i) $J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} \neq 0$ & do (ii) $u_x = u_\xi \xi_x + u_\eta \eta_x$ ...

- When F.S of $f(x)$ converges to $f(x)$:

(i) Piecewise-smooth: The interval of interest can be broken up to pieces I_i where $f(x) \in C^1(I_i)$ with $\cup I_i = \Omega_i \Rightarrow$ FS converges smth.

(ii) Matching end pts: $f(-\pi) = f(\pi)$ (i.e. $f(x_{i-1}) = f(x_i)$) $\Rightarrow$ FS converges $f(x)$.

- When Dirichlet pr-m well-posed: $u \in C^2(\Omega) \cap C^0(\bar{\Omega})$

$\boxed{\begin{aligned} \Delta u &= 0, \quad x \in \Omega \\ u(r_0, \theta) &= f(\theta), \quad \partial\Omega \end{aligned}}$

(i) Sol $\exists$ if $f(\theta) \in C^0(\partial\Omega)$

(ii) Show $\tilde{u} = u_1 - u_2$; sol-n unique

(iii) Showed u cont-ly dep. on $f(\theta)$. & sol-n

$u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right]$, a_0, a_n, b_n are F.S. coef. of $f(\theta)$.

- When Neumann Pr "well-posed". (i) $f(x) \in C^0(\partial\Omega)$

$\boxed{\begin{aligned} \Delta u &= 0, \quad \Omega \\ \frac{\partial u}{\partial n} &= f(x), \quad \partial\Omega \end{aligned}}$

(i) If sol-n exist, (ii) $\int_{\partial\Omega} f(x)\, ds = 0$

(ii) $\tilde{u} = u_1 - u_2$, see Pr $\Rightarrow \tilde{u} = C \Rightarrow$, any 2 sol-s differ by a const.

(iii) Showed sol-n dep-s cont-ly on $f(x)$. , then if $f(x) \in C^0(\partial\Omega)$ & $\int_{\partial\Omega} f(x)\, ds = 0$

exists sol-n $\int_{\partial\Omega} f\, ds = 0$ "necess-y cond-n"

Sol-n is $A_0 = 0$, sol to $\exists$.

$u(r,\theta) = \frac{A_0}{2} + \sum_{n=1}^{\infty} r^n \frac{A_n}{n} \cos(n\theta) + \frac{B_n}{n} \cdot r^n \cdot \sin(n\theta)$ where A_n, B_n = FS coef. of $f(x)$.

- When drawing or evaluating one step at a time starting with assigning ① $t = t_0$ value & ② $u(x, t_0)$ see some dif-t values of x or see where $u(x, t_0)$ gets max, ...

- Prin. part: $P_m(x, D)$ = highest partials

Prin. symb: $p_m(x, \xi)$ = highest partials replaced by ξ's

Char. eq-n: $p_m(x, \xi) = 0$. $\Rightarrow$ sol-n $\langle \xi_1, \xi_2 \ldots \xi_n \rangle$ of char. eq-n is char. dir-n.

For wave eq-n, we usually let $\xi_t^2 = 1$, & $\langle \xi_1, \xi_2 \ldots \pm 1 \rangle$'s are char. dir-ns.

- Spherical wave eq-n: $n = 3$. $\boxed{u = w/r}$ $\Longrightarrow$ turns cartesian $w_{rr} - w_{tt} = 0$.

$\boxed{\frac{1}{r^{n-1}} \cdot \frac{\partial}{\partial r}\left[r^{n-1} \cdot \frac{\partial u}{\partial r}\right] - \frac{\partial^2 u}{\partial t^2} = 0}$, this is because, $\Delta u = 0$

in spherical:

$\Delta u = \frac{1}{r^2} \cdot \frac{\partial}{\partial r}\left[r^2 \cdot \frac{\partial u}{\partial r}\right] + \frac{1}{r^2} \Lambda_3 u = 0$, where

$\Lambda_3 u = \frac{1}{\sin\varphi} \cdot \frac{\partial}{\partial \varphi}\left[\sin\varphi \cdot \frac{\partial u}{\partial \varphi}\right] + \frac{1}{\sin^2\varphi} \cdot \frac{\partial^2 u}{\partial \theta^2}$, & polar, $\varphi = \frac{\pi}{2}$, $\Lambda_3 u = \frac{\partial^2 u}{\partial \theta^2}$, we come that wave eq-n in spherical doesn't depend φ & θ, $\Lambda_3 u = 0$, giving the box formula.

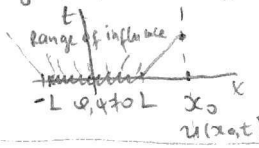
- $u_{xx} - \frac{1}{c^2} u_{tt} = 0$, $\boxed{x = c \cdot t}$ & $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t}$ & changing $\psi(x)$ works for D'Alamb

IC: $u(x,0) = \varphi(x)$, $u_t(x,0) = \psi(x)$; $\Rightarrow \varphi(x) \in C^2(\Omega)$, $\psi(x) \in C^1(\Omega)$ for sol-n $\exists$.

BC: (D): $u(\underline{x}, t) = f(\underline{x}, t)$, $\Rightarrow f \in C^2(\partial\Omega)$ for a sol-n to exist

(N): $\frac{\partial u}{\partial n} = g(\underline{x}, t)$, $\underline{x} \in \partial\Omega$, $\Rightarrow g \in C^0(\partial\Omega)$ & $\int_{\partial\Omega} g(\underline{x}, t)\, d\underline{x} = 0$ for sol-n $\exists$

- Range of influence (support from $\varphi, \psi \neq 0$):



Always consider where ($x = x_0$?) we're asking the question for $t = ?$, $u \neq 0$, or $u = 0$.

- $u_{x_1x_1} + u_{x_2x_2} - u_{tt} = 0$, $\underline{x} \in \mathbb{R}^2$, $t > 0$

IC: $u(\underline{x}, 0) = \varphi(\underline{x})$, $u_t(\underline{x}, 0) = \psi(x)$, $x \in \mathbb{R}^2$, then sol-n

$n = 2$: $u(\underline{x}, t) = \frac{1}{2\pi} \int_{B(\underline{x}, t)} \frac{\psi(x_1', x_2')\, dx_1'\, dx_2'}{\sqrt{t^2 - (x_1 - x_1')^2 - (x_2 - x_2')^2}} +$

$+ \frac{\partial}{\partial t}\left\{\underbrace{\frac{1}{2\pi} \int_{B(\underline{x}, t)} \frac{\varphi(x_1', x_2')\, dx_1'\, dx_2'}{\sqrt{\ldots}}}_{\text{entire disk.}}\right\}$,

$n = 3$, $u(\underline{x}, t) = \frac{1}{4\pi t} \int_{S(\underline{x}, t)} \psi(\underline{x}')\, dS_t + \frac{\partial}{\partial t}\left[\frac{1}{4\pi t} \int_{S(\underline{x}, t)} \varphi(\underline{x}')\, dS_t\right]$, entire ball in 3D

- Green's 1st identy: $\int_\Omega u \cdot \Delta w\, dv = \int_{\partial\Omega} u \frac{\partial w}{\partial n}\, ds - \int_\Omega \nabla u \cdot \nabla w\, dv$; [Mostly $u = w = \tilde{u}$].